Powering Prosperity

INNOVATING INDIA'S POWER SECTOR FOR A BRIGHTER, MORE SUSTAINABLE FUTURE

Q | People say that the power sector has been transformed under your leadership. There are no power shortages any more. We also see the share prices of all power sector companies going up, and investments coming in. Can you share with us the details of what you did?

We have added close to 202 GW of capacity in last 9-10 years. The current capacity is around 434 GW. Almost 45% of India's installed capacity since independence has been added in the last 10 years. In 2014 the system could meet a maximum demand of 129 GW – now the maximum demand met is 241 GW. We are now exporting power to our neighbouring countries. In FY 2023, India exported 2.4 BUs to Nepal, Bhutan and Bangladesh becoming a net exporter of power.

Q | The Prime Minister talks about one nation one grid: What is this?

In 2014, only 36 GW of power could be transferred from one part of the country to another – The grid had not reached many areas of the country. We have added about 1,91,000 ckt kms of transmission lines - connecting the whole country into one grid. This unified grid today has capacity to transfer 116 GW power from one corner of the country to another, all while maintaining a consistent frequency. India's electric grid is today the largest synchronized grid in the world.

In order to modernize transmission sector and making it smart & future ready following steps, inter-alia, have been taken:

- Control Centres have been set up to remotely monitor and operate Extra High Voltage (EHV) substations, towards the establishment of an automated and digitally controlled grid.
- Computer Security Incident Response Team in Power Sector (CSIRT-Power) has been notified for managing the cyber security incidents in Power Sector.
- Communication System in Power Sector has to be in compliance with "Central Electricity Authority (Cyber Security in Power Sector) Guidelines".
- Central Power Research Institute (CPRI) has established test facilities for cyber security for sub-station equipment.
- GIS mapping of substations and lines likely to be affected by natural disasters. The design requirements for cyclone prone areas have been changed.

Artificial Intelligence (AI) / Machine Learning (ML) tools are being used for maintenance of transmission lines. All grid substations use sensors for measuring various parameters of power system to carry out predictive maintenance.

Q | Minister, 70 years after independence, it was only under this Government the country could achieve universal access to electricity. How did you do it?

It was a tragedy that even after 70 years of independence, crores of people were still living in darkness as on 2014. Our Hon'ble PM set us a target of connecting every village and every hamlet in 1000 days. We did this in 987 days. Thereafter, we launched the Saubhagya Scheme to connect every house which did not have access to electricity. We did this in 19 months. We connected 29 million homes. The IEA called this the largest and fastest expansion of electricity access in the world.

The other problem was availability of electricity. We were adding generation and transmission capacity; but we also needed to strengthen the distribution system. We spent over 1.85 lakh crore for strengthening the distribution system under Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY), Integrated Power Development Scheme (IPDS) IPDS. Under these schemes, 2927 new sub-stations were added, 3987 existing sub-stations were upgraded, 8.35 lacs circuit kilometres of HT and LT lines were added / changed, and 6.92 lakh transformers were set-up. As a result of these, the average availability of power in rural areas which was an average of 12 hours in 2015, went upto an average of 21 hours per day today. In urban areas it is 23.78 hours.

Q | What are the steps which you have taken to improve power systems?

We have transformed the systems and processes. Some of the systems we have put in place are as follows :-

- **Economic Dispatch:** We have put in place a system whereby the cheapest power is dispatched first; therefore reducing overall system cost and hence consumer tariff. The saving due to SCED since its inception in April, 2019 till January, 2024 was around Rs. 3300 Crores.
- **Real Time Market:** The market provides an opportunity for the buyers and the sellers to buy and sell electricity closer to real time.
- We have provided for Green Day Ahead and Green Term Ahead markets.
- **Bundling of RE with Thermal electricity:** Solar is already cheaper then thermal and thus bundling RE with coal electricity will reduce power purchase cost of DISCOMs, finally benefitting the consumers.
- **Allowing flexibility to State/Central Gencos in utilizing domestic coal :** Gencos can now allocate more coal to their most efficient plants and thus generate cheaper electricity. By doing this, they also save in transportation cost. All of these finally benefits consumers.